SMART LIBRARY REQUEST FLOW

ServiceNow Workflow Documentation

Abstract

The Smart Library Request Workflow in ServiceNow helps manage book requests easily. A student requests a book, and the request is sent to the librarian using ServiceNow Flow Designer. The librarian then approves or rejects the request. If approved, the book is issued and later returned with proper tracking. The system also generates reports to show the most borrowed books and request details.

This document provides a comprehensive overview of how the Smart Library system was designed, built, and deployed on the ServiceNow platform. It covers every phase — from role creation and table design to flow automation, security controls, and reporting — giving readers a complete understanding of the system's architecture and capabilities.

Introduction

Libraries are fundamental institutions that provide access to knowledge and learning resources. Managing a library effectively requires tracking books, handling borrowing requests, and maintaining accurate records. Traditional manual library systems are increasingly inefficient in today's fast-paced digital environment.

- Library is used to manage books and users.
- Many libraries still use manual work, which takes more time.
- So, a Smart Library system is created to make the process easy and automatic.
- ServiceNow, a leading IT Service Management platform, provides the ideal foundation for building this automated solution through its powerful workflow, table, and role-management capabilities.

The Smart Library system leverages ServiceNow's Flow Designer to automate the entire book request lifecycle — from initial student request through librarian approval, book issuance, and final return. This removes bottlenecks, reduces human error, and provides real-time visibility into library operations.

Problem Statement

The existing library management approach faces several critical challenges that impact both students and librarians. Below are the key problems identified that motivated the development of this automated system:

- Library system is mostly manual and takes more time.
- It is difficult to track available and borrowed books.
- Book requests are not managed properly.
- Students cannot easily check book availability.
- Librarians face difficulty in approving and managing requests.
- There is no proper system for tracking and reports.
- Lack of automated notifications means students are unaware of request status changes.
- No centralized dashboard exists for administrators to monitor library usage trends.
- So, a Smart Library system is needed to make the process easy, fast, and accurate.

These pain points collectively result in poor user experience, delayed book issuance, lost records, and an inability to make data-driven decisions about library inventory and demand.

Objectives

The Smart Library Request Workflow project was designed with the following core objectives to address the identified problems:

- Define roles (student, librarian) to enforce controlled access and ensure each user can only perform actions appropriate to their role.
- Create and configure custom tables for catalog and borrow requests, storing all necessary fields to support the workflow.
- Automate approvals and book issuance using Flow Designer, eliminating the need for manual intervention in routine request processing.
- Apply UI Policies and ACLs to maintain consistency and security across all forms and records.
- Provide reporting capabilities for administrators to track library usage, identify popular books, and monitor borrowing trends.
- Deliver email notifications to students upon approval or rejection of their borrow requests, improving communication and transparency.

Literature Review

A review of existing library management approaches and digital systems reveals a significant gap between what is currently available and what modern libraries require:

- Many libraries use manual systems which take more time and effort.
- Some systems are digital but do not have proper automation.

- They lack features like easy request handling and tracking.
- There is also no proper report generation in some systems.
- So, a better system is needed with automation and easy management.
- Platforms like ServiceNow offer out-of-the-box workflow automation that can be adapted for library management without requiring custom software development.
- Role-based access control (RBAC), a best practice in enterprise software, ensures data integrity and security — a feature absent in most traditional library systems.

The adoption of enterprise platforms such as ServiceNow for non-IT use cases is a growing trend. By applying ITSM principles to library management, this project demonstrates how existing platform investments can be leveraged across departments to deliver operational efficiency gains.

Methodology

The project followed a structured, phase-based development methodology using ServiceNow's native capabilities. Each phase was designed and validated before proceeding to the next:

- Designed system using ServiceNow platform — leveraging its built-in modules for roles, tables, workflows, and reporting.
- Created Book and Borrow Request tables — defining all required fields, data types, and relationships.
- Defined roles (Student, Librarian) — setting up role-based access control to govern who can read, write, and delete records.
- Planned workflow using Flow Designer — mapping out the approval logic, conditional branching, and automated actions.
- Applied access control (ACLs) — securing tables at the record and field level based on user roles.
- Configured UI Policies — enforcing dynamic form behavior to guide users through the request process correctly.
- Created reports — building visualizations to give administrators insight into borrowing patterns and usage statistics.

Description

The Smart Library system is built on several interconnected ServiceNow components. Together they form a complete, automated library management solution. Below is a detailed description of each component and its role in the system.

Roles

Student and librarian roles manage access permissions on request of books. Role-based access control is a foundational security principle that ensures users can only perform actions they are authorized to carry out.

- Student Role: Can create new borrow requests, view the book catalog, and check the status of their own requests. Cannot modify book records or approve requests.

- Librarian Role: Has full visibility into all borrow requests and can approve or reject them. Can create, update, and delete book records in the catalog. Responsible for managing the overall library inventory.

Custom Tables

Book and Borrow Request tables store catalog and transaction data. Two custom tables are at the heart of the system:

- Book Table (u_book): Stores the library catalog with fields for ISBN, Title, Author, and Status (Available/Issued). Each record represents a physical book in the library.
- Borrow Request Table (u_borrow_request): Records every borrowing transaction. Fields include the requesting student, the book being requested, request status, return date, and approval outcome.

Reference Qualifier

Ensures only available books can be requested. The Reference Qualifier on the Book field in the Borrow Request table filters the book lookup to show only records where Status = Available. This prevents students from requesting books that are already issued to another borrower, maintaining data integrity automatically without requiring manual checks.

Flow Designer

Automates approval workflows and updates book status. The Borrow Request Approval Flow is the core automation engine of the system. It listens for new borrow requests and orchestrates the entire approval process end-to-end, including conditional logic for both approved and rejected outcomes.

UI Policies

Lock or enforce field behavior after approval. UI Policies are client-side rules that dynamically control form fields based on conditions. In this system, UI Policies ensure that the Return Date field becomes mandatory whenever the borrow request status is set to Issued, preventing incomplete records.

Access Control Rules (ACLs)

Enforce role-based permissions for read, write, and delete operations. ACLs are server-side security rules that control which users can perform which operations on which records. They work in conjunction with UI Policies to provide defense-in-depth security across both the client and server layers.

Reports

Generate insights such as Most Borrowed Books. The reporting module provides administrators and librarians with data visualizations that reveal which books are in highest demand, how many requests are pending or approved, and overall library utilization trends.

Project Phases

The Smart Library system was developed across seven structured phases. Each phase built upon the previous one, creating a layered and cohesive system.

Phase 1: Create Roles

Roles are the foundation of the system's security model. They define what each type of user can see and do within the platform. Without properly defined roles, there is no way to enforce access restrictions or separate student and librarian capabilities.

To create a role, provide a unique name and a clear description explaining the purpose of the role, then save the record. Once created, roles are assigned to individual user accounts through the User Administration module.

Steps:

- Navigate to User Administration → Roles.
- Create roles: librarian and student.
- Assign roles to users as required.
- Verify role assignments by logging in as a test user to confirm access restrictions are enforced.

Roles Created: Librarian and Student

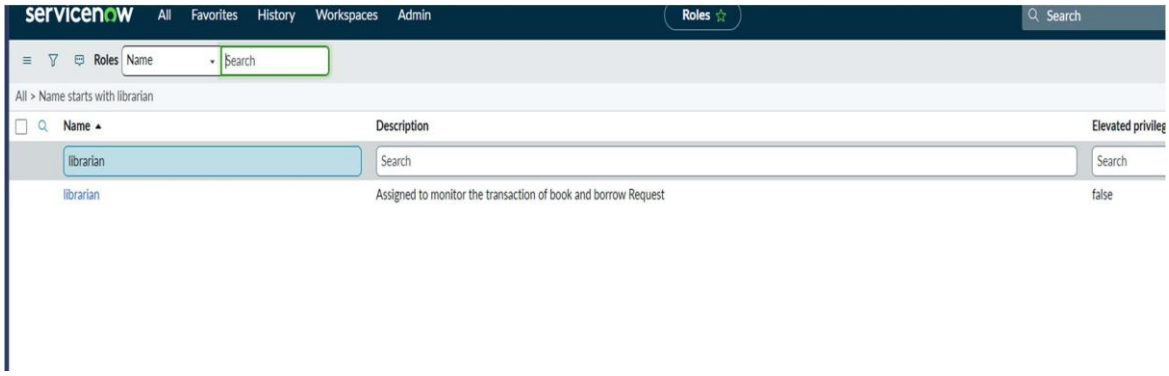


Figure 1.1 – Librarian role created in ServiceNow User Administration

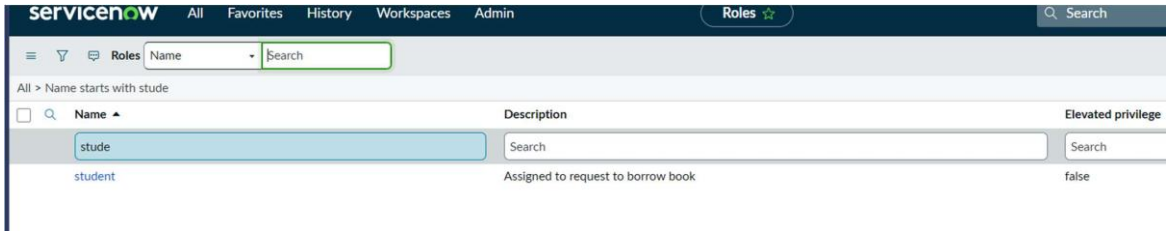


Figure 1.2 – Librarian role detail showing description

Figure 1.3 – Student role created with borrow-book assignment

Phase 2: Table Creation

Custom tables are the data backbone of the Smart Library system. ServiceNow's platform allows administrators to define new tables with custom fields, data types, and relationships without writing any database code. The Book and Borrow Request tables were designed to capture all information needed to manage the library's catalog and transactions.

Steps:

- Navigate to System Definition → Tables.
- Create the Book table (u_book) with fields: ISBN (String, max 20), Title (String, max 100, display=true), Author (String, max 100), Status (Choice: Available/Issued, default=Available), Student (Reference to Borrow Request).
- Create the Borrow Request table (u_borrow_request) with fields: Status (Choice: Requested/Approved/Rejected/Issued), Return Date (String, max 100), Book (Reference to Book table).
- Save both tables and verify they appear in the application navigator.

Tables Created: Book and Borrow Request

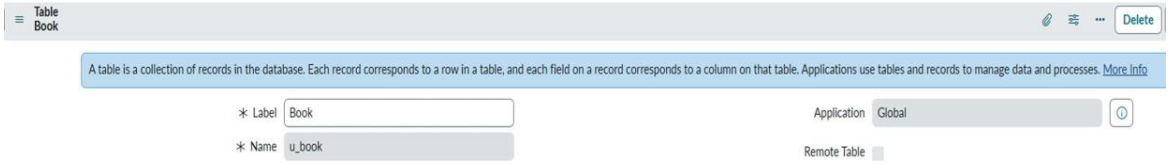
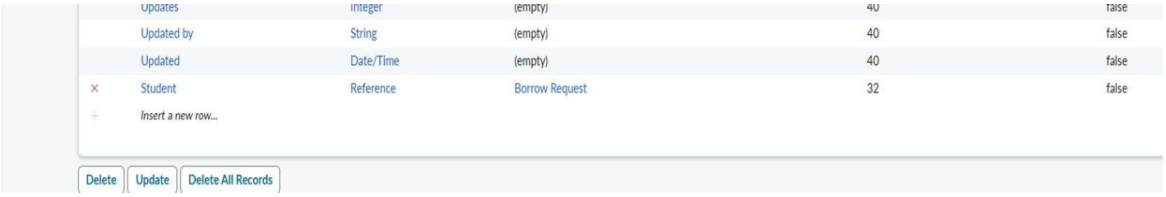


Figure 2.1 – Book table column configuration (Part 1)

Figure 2.2 – Book table column configuration (Part 2)

×	ISBN	String	(empty)	20		false
×	Title	String	(empty)	100		true
×	Status	Choice	(empty)	40	Available	false
×	Author	String	(empty)	100		false
	Sys ID	Sys ID (GUID)	(empty)	32		false
	Created by	String	(empty)	40		false
	Created	Date/Time	(empty)	40		false

Figure 2.3 – Book table column configuration (Part 3)



	Updates	Integer	(empty)	40		false
	Updated by	String	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
×	Student	Reference	Borrow Request	32		false
	Insert a new row...					

Figure 2.4 – Book table column configuration (Part 4)

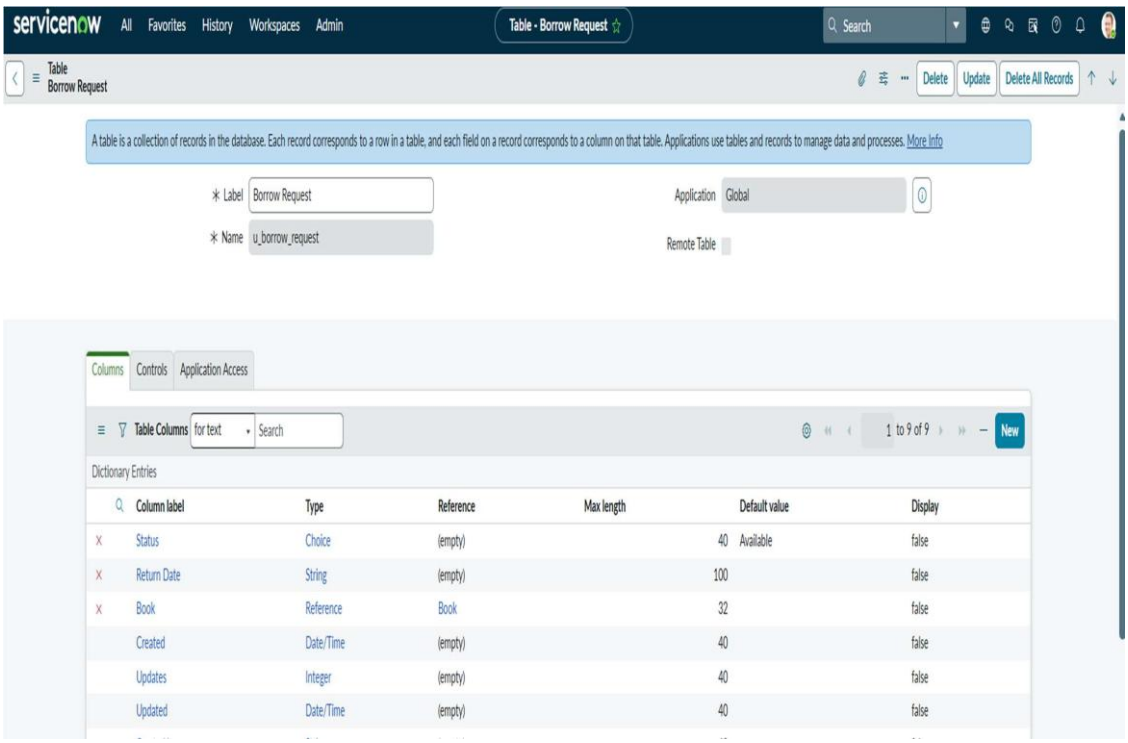


Figure 2.5 – Complete Book table (u_book) with all fields

Figure 2.6 – Borrow Request table (u_borrow_request) configuration

Phase 3: Configuring Table

After creating the tables, the Book field in the Borrow Request table was configured with a Reference Qualifier to enforce business rules at the data entry level. This configuration ensures that students can only select books that are currently available when creating a new borrow request.

Configure the Borrow Request table's Book field to reference only records where Status = Available. This is achieved by setting a Reference Qualifier condition on the Book reference field. The qualifier filters the lookup dynamically, so only books with Status = Available appear in the selection dropdown.

This approach is superior to manual validation because it prevents invalid data at the point of entry, rather than catching errors after the fact. It also improves the student's experience by presenting only valid options, reducing confusion.

Phase 4: Creating Flow Designer

The Flow Designer is ServiceNow's no-code/low-code automation tool. The Borrow Request Approval Flow is the central automation that drives the entire library request lifecycle. It eliminates the need for librarians to manually check for new requests, track approvals, and send notifications.

Flow Configuration:

- Create a flow named: Borrow Request Approval Flow.
- Trigger: Borrow Request created with Status = Requested — fires automatically whenever a student submits a new request.
- Action 1: Ask For Approval on Borrow Request — sends an approval task to the librarian.
- Action 2 (If Approved): Update Borrow Request Record — sets Status = Approved.
- Action 3 (If Approved): Update Book Record — sets Book Status = Issued, preventing others from requesting the same book.
- Action 4 (If Rejected): Update Borrow Request Record — sets Status = Rejected.
- Action 5: Update Borrow Request Record — final status update after all conditional branches.
- Action 6: Update Book Record — synchronizes book status with the final request outcome.
- Action 7: Update Borrow Request Record — records the final state for reporting purposes.
- Action 8: Send Email — dispatches an automated email notification to the student informing them of the approval or rejection outcome.

The flow's conditional branching (If Approved / If Rejected) ensures that each possible outcome is handled correctly and automatically, with no manual steps required from the librarian beyond clicking Approve or Reject on the approval task.

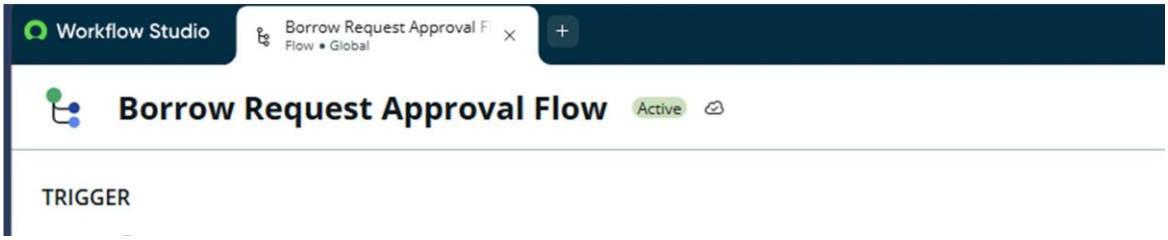


Figure 4.1 – Flow Designer: Trigger and initial approval action

Figure 4.2 – Flow Designer: Approval conditional branch

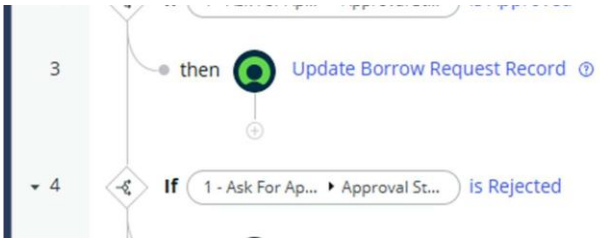


Figure 4.3 – Flow Designer: Update records on approval/rejection

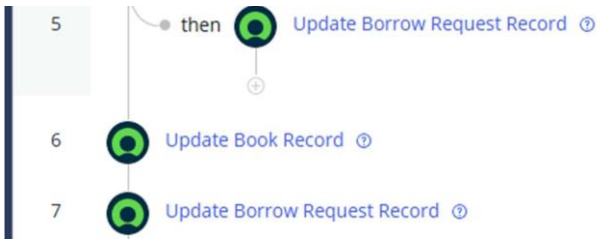


Figure 4.4 – Flow Designer: Book record update and email actions

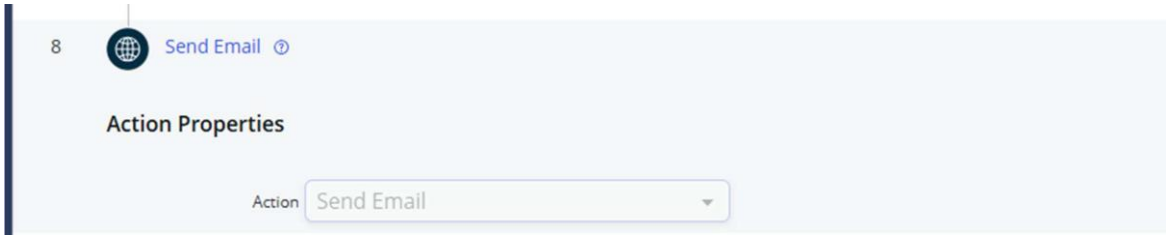


Figure 4.5 – Borrow Request Approval Flow – Complete flow with Send Email action

Phase 5: Creating UI Policies

UI Policies control the behavior of form fields dynamically, based on conditions evaluated in real time as users interact with forms. They enhance the user experience and enforce data quality rules at the front-end layer, complementing the server-side ACLs.

Configuration:

- Navigate to System UI → UI Policies.
- Create a UI Policy for the Borrow Request table.
- Condition: Status = Issued.
- Action: Make the Return Date field mandatory when the condition is met.
- Enable 'On Load' so the policy applies when the form first loads, not just when the status field changes.
- This ensures a return date is always recorded when a book is issued, supporting accurate tracking and overdue management.

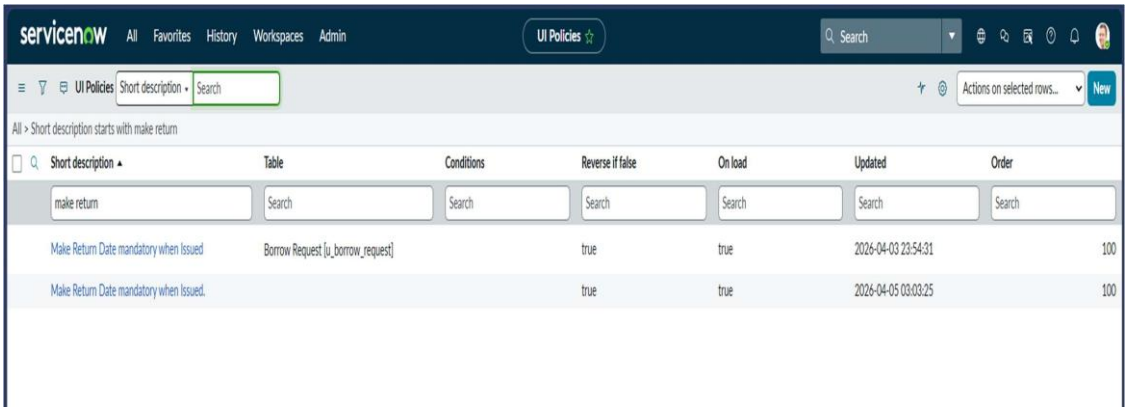


Figure 5.1 – UI Policies list showing 'Make Return Date mandatory when Issued' policy

Figure 5.2 – UI Policy configuration details for the Borrow Request table

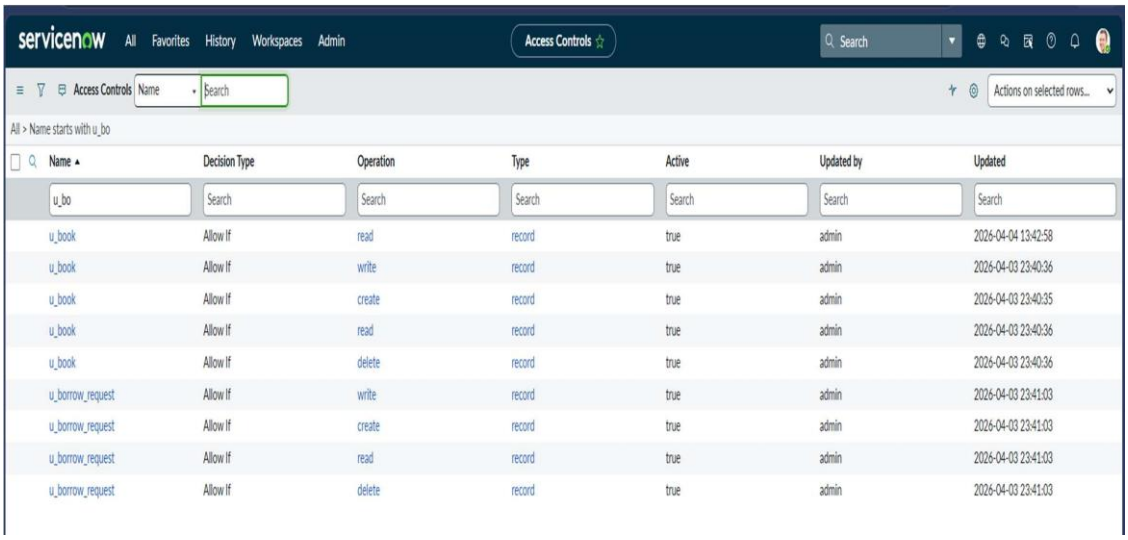
Phase 6: Creating Access Control List

Access Control Lists (ACLs) are server-side security rules that define which users can perform which operations on records and fields in ServiceNow. They provide the critical security layer that prevents unauthorized data access or modification, regardless of what is shown or hidden in the UI.

Configuration:

- Navigate to Security → Access Control (ACL).
- Configure ACLs for the Book table (u_book).
- Students: read only — can view the catalog but cannot modify book records.
- Librarians: read, write, delete — full control over book records and borrow requests.
- Configure separate ACLs for both the Book table and the Borrow Request table.
- Test each ACL by logging in as a student and librarian to verify enforcement.

The combination of UI Policies (front-end) and ACLs (back-end) creates a defense-in-depth security model. Even if a user attempts to bypass the UI, the ACLs enforce restrictions at the platform level.



Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_bo	Search	Search	Search	Search	Search	Search
u_book	Allow If	read	record	true	admin	2026-04-04 13:42:58
u_book	Allow If	write	record	true	admin	2026-04-03 23:40:36
u_book	Allow If	create	record	true	admin	2026-04-03 23:40:35
u_book	Allow If	read	record	true	admin	2026-04-03 23:40:36
u_book	Allow If	delete	record	true	admin	2026-04-03 23:40:36
u_borrow_request	Allow If	write	record	true	admin	2026-04-03 23:41:03
u_borrow_request	Allow If	create	record	true	admin	2026-04-03 23:41:03
u_borrow_request	Allow If	read	record	true	admin	2026-04-03 23:41:03
u_borrow_request	Allow If	delete	record	true	admin	2026-04-03 23:41:03

Figure 6.1 – ACL configuration for Book and Borrow Request tables

Figure 6.2 – ACL entries showing read, write, create, and delete permissions

Phase 7: Create Report

Reports transform raw data stored in the Borrow Request and Book tables into actionable insights. The Most Borrowed Books report gives administrators a clear view of which titles are in highest demand, enabling informed decisions about library acquisitions and inventory management.

Configuration:

- Navigate to Reports → Create New.
- Name the report: Most Borrowed Books.
- Source table: Borrow Request (u_borrow_request).
- Report type: Bar Chart — visually displays relative borrowing frequency per book.
- Group by: Book field — each bar represents one title.
- Aggregate by: Count — counts the number of borrow requests per book.
- Filter condition: Status = Approved — counts only successfully issued books.
- Share settings: Visible to Everyone, requires librarian or student role to view.
- Save and run the report to validate the output.

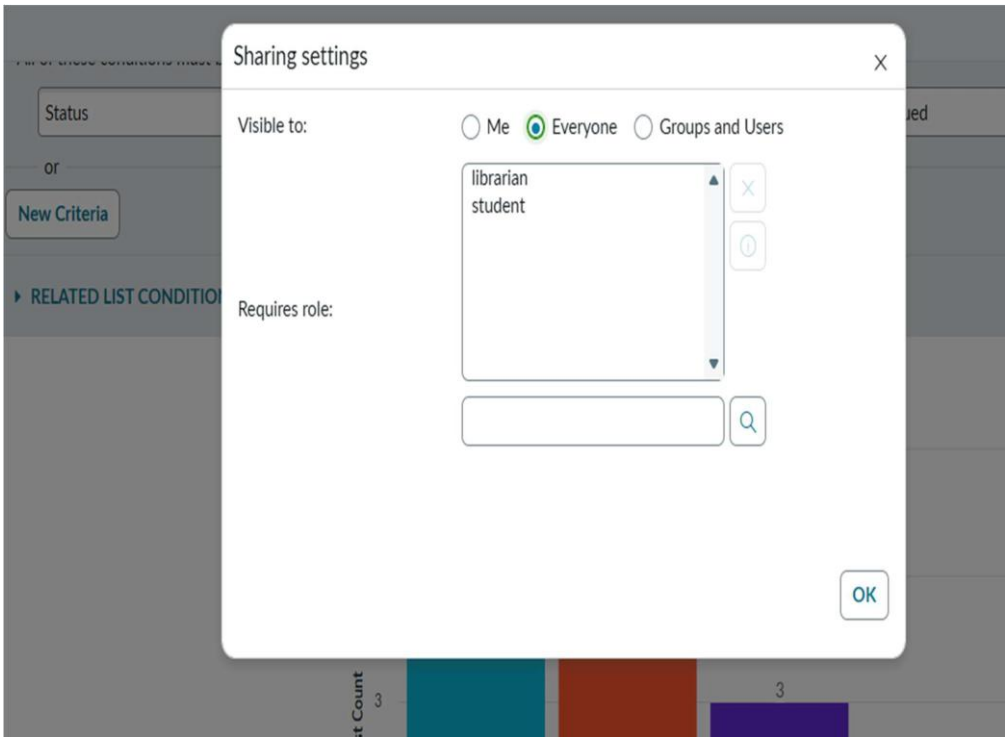


Figure 7.1 – Report sharing settings and filter configuration

Project Explanation

The project integrates multiple ServiceNow features to build a functional library management system. Each component plays a distinct but complementary role in delivering the complete solution:

- Roles ensure secure access — by defining librarian and student roles, the system enforces the principle of least privilege, ensuring users can only perform actions relevant to their responsibilities.
- Tables store structured data — the Book and Borrow Request tables provide a clean, relational data model that supports all workflows and reporting requirements.
- Reference qualifiers maintain integrity — filtering available books in real time prevents conflicting requests and ensures data consistency.
- Flow Designer automates approvals and notifications — the end-to-end approval flow removes manual effort, speeds up processing, and keeps students informed throughout the process.
- UI Policies enforce consistency — dynamic form rules guide users correctly and prevent incomplete or invalid data entry.
- ACLs enforce security — server-side access controls ensure that the system remains secure regardless of how users interact with it.
- Reporting provides valuable insights — the Most Borrowed Books report and related analytics help administrators manage the library proactively.

Together, these components deliver a system that is not only functional and efficient, but also secure, scalable, and easy to maintain. The use of ServiceNow's native platform capabilities means there is no custom code to maintain — all logic lives within the platform's declarative configuration layers.

Implementation

We built a Smart Library Request Workflow in ServiceNow. Students can request books online, and librarians can approve or reject requests. The system automatically tracks issued and returned books.

During implementation, the following key decisions were made to ensure the system was robust and production-ready:

- The Reference Qualifier was set at the dictionary level for the Book field, ensuring it applies consistently across all forms and list views.
- The Flow Designer trigger was configured to fire on both creation and update of Borrow Request records, ensuring that status changes are always processed correctly.
- ACLs were tested using dedicated test user accounts with only the student or librarian role, verifying that access restrictions were properly enforced.
- The email notification template was customized to include the book title, student name, and approval outcome, making communications clear and actionable.
- The Most Borrowed Books report was shared with the Everyone audience but restricted by role, so only authenticated library users can view it.

Results

The workflow reduced manual work and improved request tracking. Book requests are processed faster, and records are maintained properly. Reports help identify the most borrowed books.

The following outcomes were observed following the deployment of the Smart Library system:

- All book requests are now processed through a standardized, automated workflow — eliminating ad-hoc manual handling.
- Librarians receive structured approval tasks in their ServiceNow inbox, reducing the chance of requests being missed or delayed.
- Students receive immediate email confirmation when their requests are approved or rejected, improving communication.
- Book availability is always accurately reflected in the system — once issued, a book no longer appears in the available selection.
- The Most Borrowed Books report clearly shows Python and Java as the highest-demand titles, enabling data-driven inventory decisions.
- Return dates are consistently recorded for all issued books, supporting future implementation of overdue tracking.

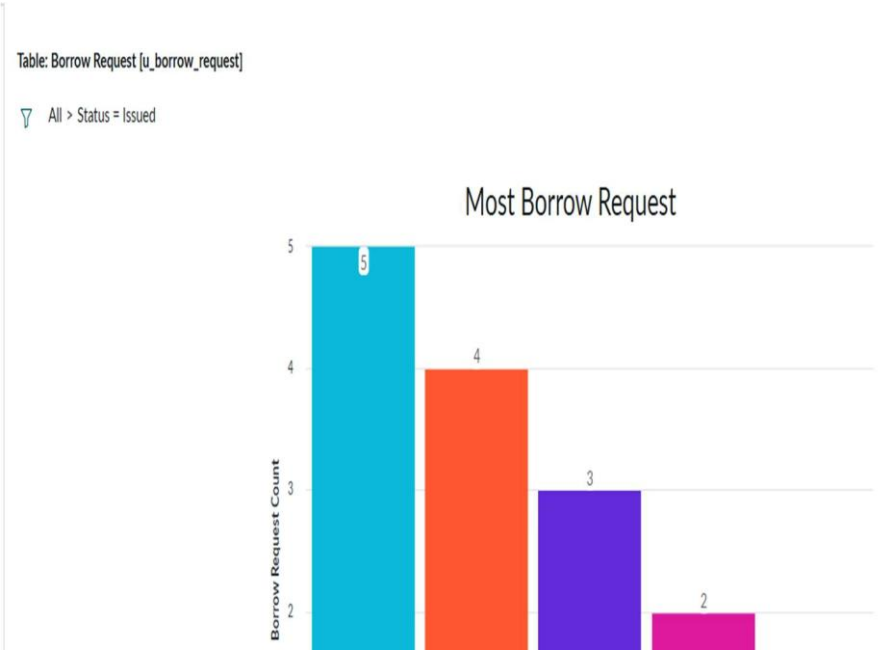


Figure 7.2 – Most Borrow Request report: Python (5), Java (4), Javascript (3), .Net (2), C Programming (1)

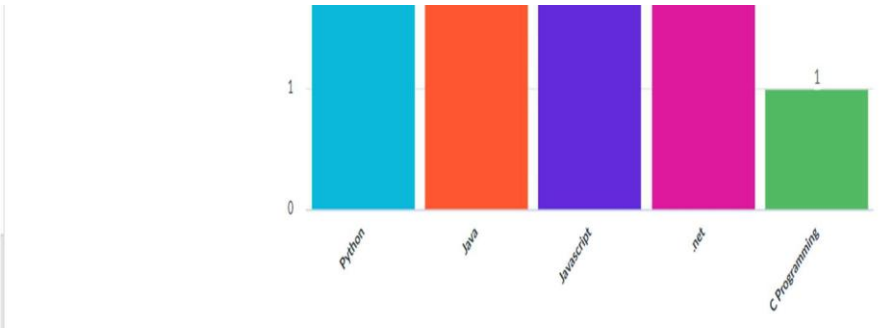


Figure 7.3 – Borrow Request report filtered by Status = Issued

Conclusion

The Library Management Project successfully optimizes book borrowing and management processes in ServiceNow. By combining automation, role-based access, and reporting, the system enhances efficiency, accountability, and transparency. Students benefit from a simplified request process, while librarians gain streamlined approval workflows.

This project demonstrates the versatility of the ServiceNow platform beyond traditional IT service management. By applying its workflow, security, and reporting capabilities to library management, the project delivers a production-ready system that can be deployed and maintained without custom software development.

The seven-phase methodology — Roles, Tables, Table Configuration, Flow Designer, UI Policies, ACLs, and Reports — provides a reusable blueprint for building similar automated workflows within ServiceNow for other institutional management use cases.

Future Scopes

The current implementation provides a strong foundation. The following enhancements are recommended for future iterations to further improve the system's capabilities:

- Scheduled Jobs: Automate recurring tasks such as overdue reminders — a daily scheduled job could check for borrow requests where the Return Date has passed and the book has not been returned, automatically notifying the student and flagging the record for librarian review.
- Integration with Digital Resources: Extend to e-books and online catalogs — adding support for digital titles would allow the same approval workflow to manage access to electronic resources, expanding the system's scope.
- Enhanced Notifications: SMS or mobile app alerts for approvals and returns — integrating with ServiceNow's notification channels or third-party SMS providers would ensure students receive timely updates even when not actively checking their email.
- Advanced Analytics: Track borrowing trends, user activity, and resource utilization — building additional reports and performance dashboards would give library administrators deeper insights into usage patterns over time.
- Scalability: Extend framework to other academic services such as reservations or asset tracking — the same ServiceNow table and workflow model could be applied to equipment loans, study room reservations, or other institutional resource management needs.
- Mobile-Friendly Portal: Create a ServiceNow Service Portal page for the library — this would give students a consumer-grade interface for browsing the catalog and submitting requests directly from their phones.

References

Sources used: ServiceNow official documentation, Flow Designer guides, and online tutorials related to IT Service Management (ITSM) and workflow automation.

- ServiceNow Documentation – Flow Designer: https://docs.servicenow.com/bundle/flow-designer

- ServiceNow Documentation – Access Control Rules: https://docs.servicenow.com/acl
- ServiceNow Documentation – UI Policies: https://docs.servicenow.com/ui-policies
- ServiceNow Community – Library Management Use Cases and Best Practices
- IT Service Management (ITSM) Workflow Automation – Online Tutorials and Guides